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Lab 07
gm/ID Design Methodology

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PART 1: OTA Design

1. Design Specifications

Parameter	Target Specification (0.18um CMOS)
Supply Voltage	1.8V
Load Capacitance (C_L)	5pF
Open Loop DC Voltage Gain	$\geq 34\text{dB}$
CMRR @ DC	$\geq 74\text{dB}$
Phase Margin	≥ 70
OTA Current Consumption	$\leq 20\mu\text{A}$
CM Input Range (Low)	$\leq 0.8\text{V}$
CM Input Range (High)	$\geq 1.5\text{V}$
Gain-Bandwidth Product (GBW)	$\geq 5\text{MHz}$

2. Design Methodology and Hand Calculations

1. Topology Selection

Based on the design specifications, the required Common-Mode Input Range (CMIR) extends towards the supply voltage (V_{DD}), with a high limit of at least 1.5V. To accommodate this high input common-mode voltage, an NMOS input pair topology was selected for the 5T OTA.

2. Input Pair Design (M1, M2)

The sizing of the input pair is primarily driven by the Gain-Bandwidth Product (GBW) and the open-loop DC voltage gain specifications.



A. Transconductance and Gain Requirements:

The target DC gain is $\geq 34\text{dB}$, which corresponds to a linear gain of approximately 50 V/V. Assuming the gain is given by $A_v = \frac{g_m}{2g_{ds}}$, the intrinsic gain per transistor (g_m/g_{ds}) must satisfy:

$$g_m/g_{ds} \geq 100$$

For a load capacitance $C_L = 5\text{pF}$ and a target $GBW \geq 5\text{MHz}$, the required transconductance (g_m) is calculated as follows:

$$g_m = 2\pi \times GBW \times C_L$$

$$g_m \geq 2\pi \times 5\text{MHz} \times 5\text{pF} \approx 157\mu\text{S}$$

To ensure the specifications are met with a safe margin, a transconductance of $g_m = 160\mu\text{S}$ was chosen. Given the branch current $I_D = 10\mu\text{A}$, the required g_m/I_D ratio is:

$$g_m/I_D = \frac{160\mu\text{S}}{10\mu\text{A}} = 16\text{ V}^{-1}$$

B. Sizing via ADT (Analog Designer's Toolbox):

Using the ADT, a parameter sweep for the channel length (L) was performed on the g_m/g_{ds} vs. g_m/I_D chart to select an appropriate length that satisfies $g_m/g_{ds} \geq 100$ at $g_m/I_D = 16\text{ V}^{-1}$.

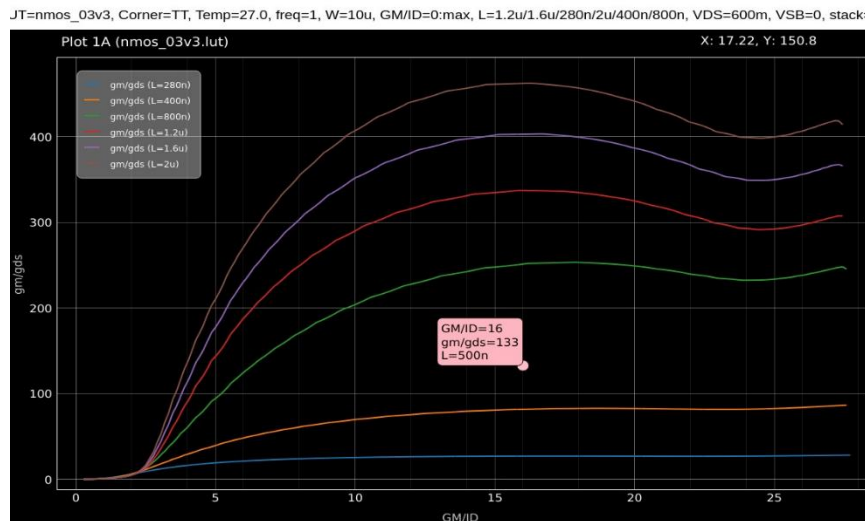


Figure 1: gm/g_{ds} vs. gm/ID



From the chart, a channel length of $L = 0.5\mu m$ was selected ($g_m/g_{ds} = 133$). Next, the current density (I_D/W) was extracted from the I_D/W vs. g_m/I_D chart for the selected L and g_m/I_D :

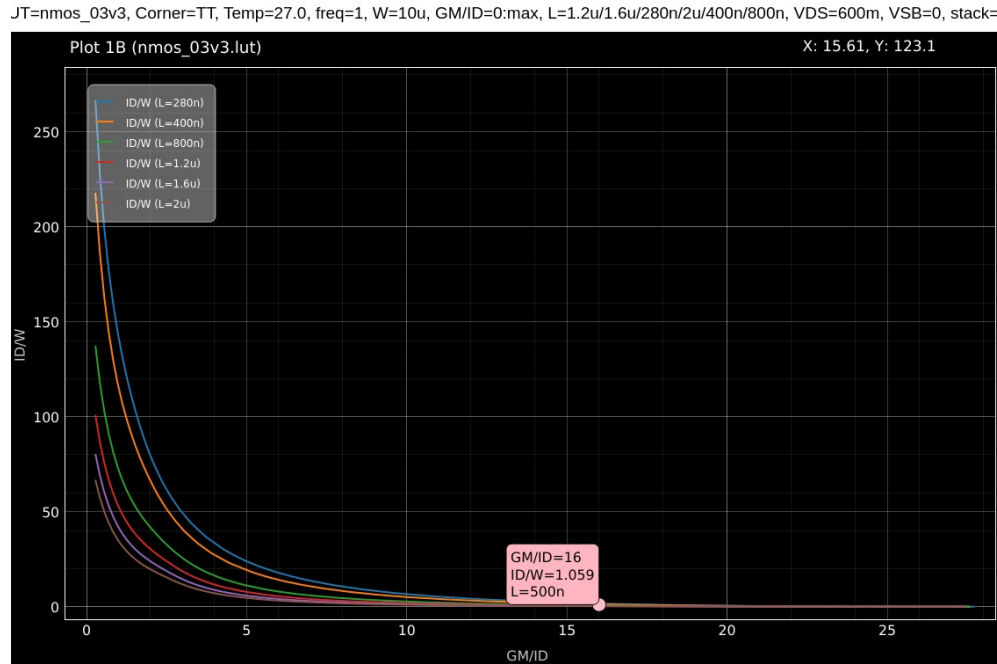


Figure 2: I_D/W vs. g_m/I_D

The chart yields a current density of $I_D/W = 1.056$ A/m. Finally, the width (W) of the input transistors is calculated as:

$$W = \frac{I_D}{I_D/W} = \frac{10\mu A}{1.056 \text{ A/m}} = 9.47\mu m$$



3. Active Load Design (M3, M4)

The active load is designed using a PMOS current mirror. The sizing is governed by the upper limit of the Common-Mode Input Range ($V_{ICM_{high}}$) and the requirement for a high output resistance to maintain the target voltage gain, while keeping transconductance relatively low to minimize noise and phase margin degradation.

A. $V_{ICM_{high}}$ Constraint and g_m/I_D Selection:

To satisfy the upper common-mode limit ($V_{ICM_{high}} \geq 1.5V$), the saturation voltage of the PMOS load ($|V_{DSAT3}|$) must be restricted. Using the simplified relationship constraint for the input pair entering the triode region, we approximate the maximum allowable $|V_{DSAT3}| \approx 0.3V$.

Using the first-order approximation $g_m/I_D \approx \frac{2}{|V_{DSAT}|}$, the minimum required ratio is:

$$(g_m/I_D)_3 \geq \frac{2}{0.3V} \approx 6.67 \text{ V}^{-1}$$

To achieve a high output resistance (r_o) for accurate current mirroring and to satisfy the overall gain requirement, a higher g_m/I_D ratio is preferred. However, it should not be excessively high to avoid unnecessarily increasing g_{m3} , which would push the mirror pole to lower frequencies and degrade the phase margin. Therefore, a balanced value of $(g_m/I_D)_3 = 15 \text{ V}^{-1}$ was selected.

Given the branch current $I_{D3} = 10\mu A$, this yields a transconductance of $g_{m3} = 150\mu S$.

B. Sizing via ADT:

To maintain the required open-loop gain, the intrinsic gain of the load devices (g_{m3}/g_{ds3}) must be comparable to the input pair requirements. The target intrinsic gain was set to approximately 124.7. By sweeping the channel length (L) on the g_m/I_D vs. g_m/g_{ds} chart for PMOS devices:

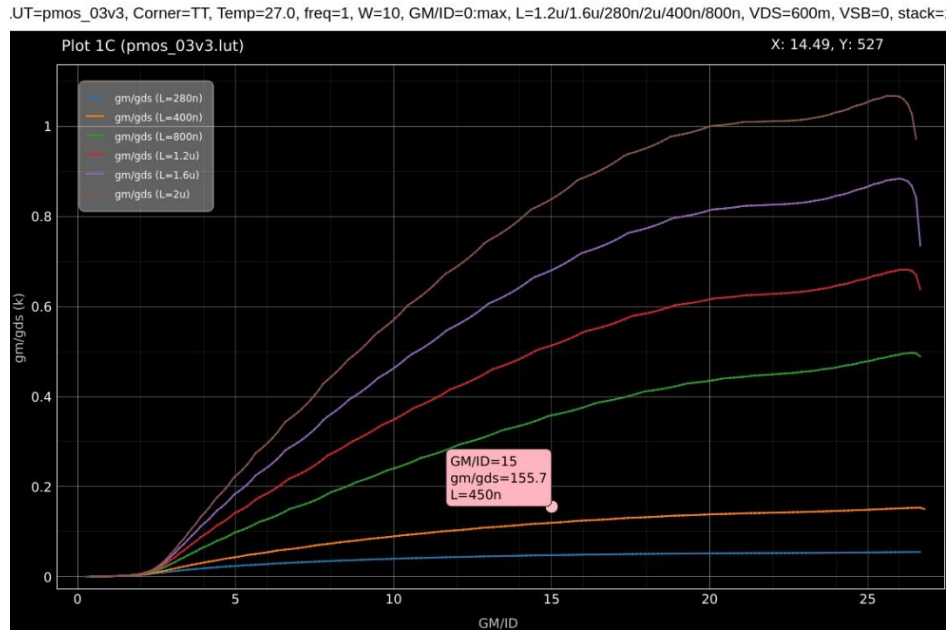


Figure 3: gm/gds vs gm/id for CM load

A channel length of $L = 0.45\mu m$ was selected to meet this condition. Next, the current density (I_D/W) was extracted from the corresponding ADT chart at $(g_m/I_D)_3 = 15 V^{-1}$:

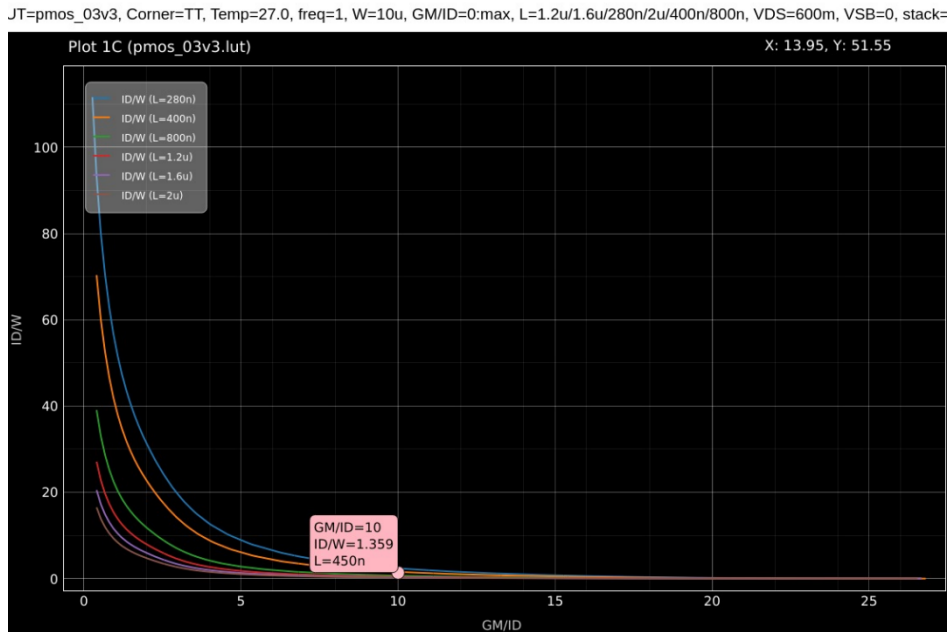


Figure 4: Id/W vs. gm/Id for CM load



The chart yields a current density of $I_D/W = 1.359 \text{ A/m}$ (or $\mu\text{A}/\mu\text{m}$). The required width (W) for the PMOS load transistors is then calculated as:

$$W_{3,4} = \frac{I_{D3}}{I_D/W} = \frac{10\mu\text{A}}{1.359 \text{ A/m}} \approx 7.4\mu\text{m}$$

4. Tail Current Source Design (M5)

The tail current source (M5) provides the total bias current ($I_{tail} = 20\mu\text{A}$). Its output resistance (R_{SS}) is a critical parameter for suppressing common-mode signals and achieving the target Common-Mode Rejection Ratio (CMRR).

A. Output Resistance (R_{SS}) Requirement:

Given the target differential gain $A_{vd} \geq 34\text{dB}$ and the required $CMRR \geq 74\text{dB}$, the maximum allowable common-mode gain (A_{vcm}) in decibels is calculated as:

$$A_{vcm(dB)} = A_{vd(dB)} - CMRR_{dB} = 34\text{dB} - 74\text{dB} = -40\text{dB}$$

Converting this to a linear scale:

$$|A_{vcm}| = 10^{-40/20} = 0.01 \text{ V/V}$$

The common-mode gain of the OTA is approximated by the following relationship:

$$|A_{vcm}| \approx \frac{1}{2g_{m3}R_{SS}}$$

Substituting the previously designed transconductance of the active load ($g_{m3} = 150\mu\text{S}$), the minimum required tail resistance is:

$$R_{SS} \geq \frac{1}{2 \times 150\mu\text{S} \times 0.01} = 333.33 \text{ k}\Omega$$

B. Sizing via ADT:

To achieve this high output resistance (for an ideal current mirror operation and superior common-mode rejection), a relatively large channel length is required.

Using the ADT, a length of $L = 1.2\mu\text{m}$ was selected. As verified from the g_m/g_{ds} vs. g_m/I_D chart, this length yields a high intrinsic gain of $g_m/g_{ds} = 334.9$, which correlates to a large r_o that easily satisfies the $R_{SS} \geq 333.33 \text{ k}\Omega$ condition.

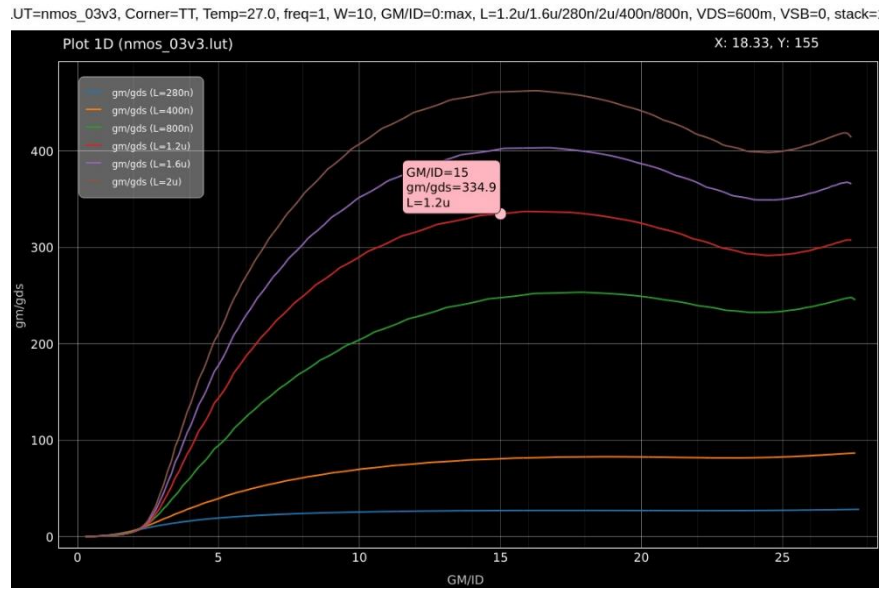


Figure 5: gm/gds vs. gm/Id for tail current

From the current density (I_D/W) chart at the selected operating point, the extracted value is $I_D/W = 0.5766$ A/m (or $\mu A/\mu m$). Given the total tail current $I_{tail} = 20\mu A$, the required width (W) for the tail current source is calculated as:

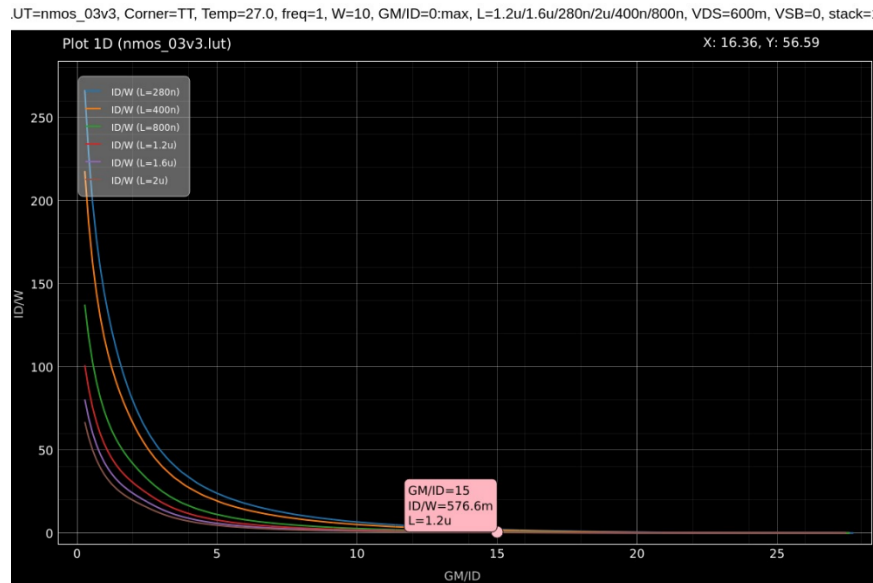


Figure 6: Id/w vs. gm/Id for tail current source

$$W_5 = \frac{I_{tail}}{I_D/W} = \frac{20\mu A}{0.5766 \text{ A/m}} \approx 34.69\mu m$$



5. Final Sizing Summary Table

Based on the g_m/I_D methodology and ADT extractions, the final dimensions and operating parameters for all transistors in the 5T OTA are summarized in the table below:

Transistor	Type	g_m/I_D (V^{-1})	L (μm)	W (μm) total
M0, M1 (Input Pair)	NMOS	16	0.5	9.47
M2, M3 (Active Load)	PMOS	<15	0.45	7.4
M4 (Tail Current)	NMOS	16.745	1.2	34.87

Note: In the Cadence Virtuoso schematic, the total width W can be implemented using multiple fingers or multipliers to optimize the layout and matching, maintaining the same total W/L ratio.

PART 2: Open-Loop OTA Simulation

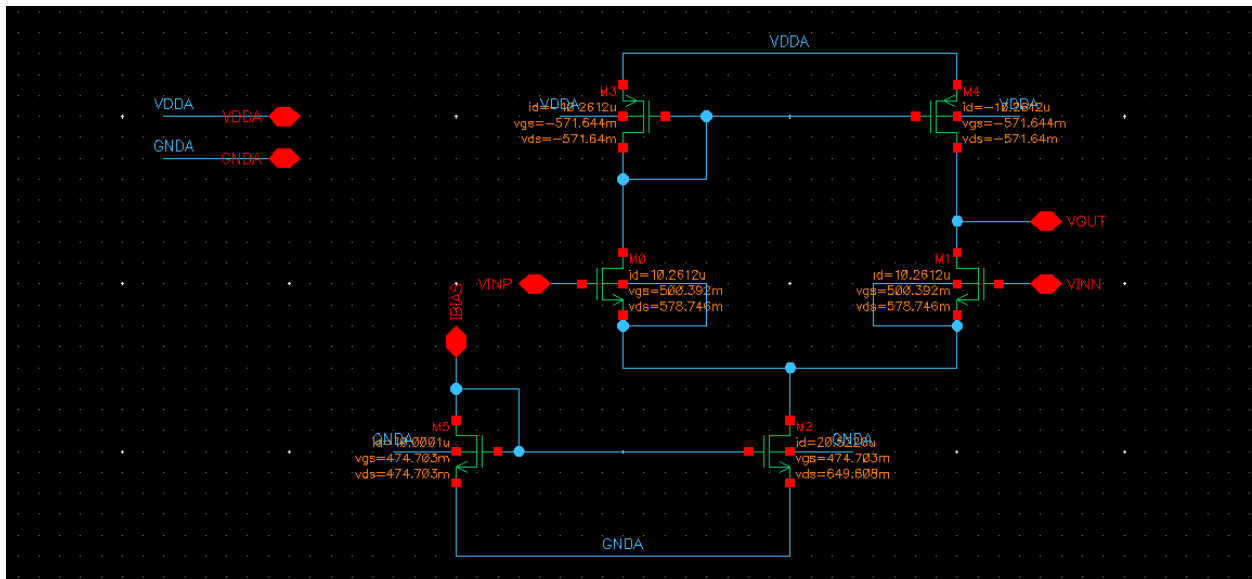


Figure 7: OTA-5t schematic

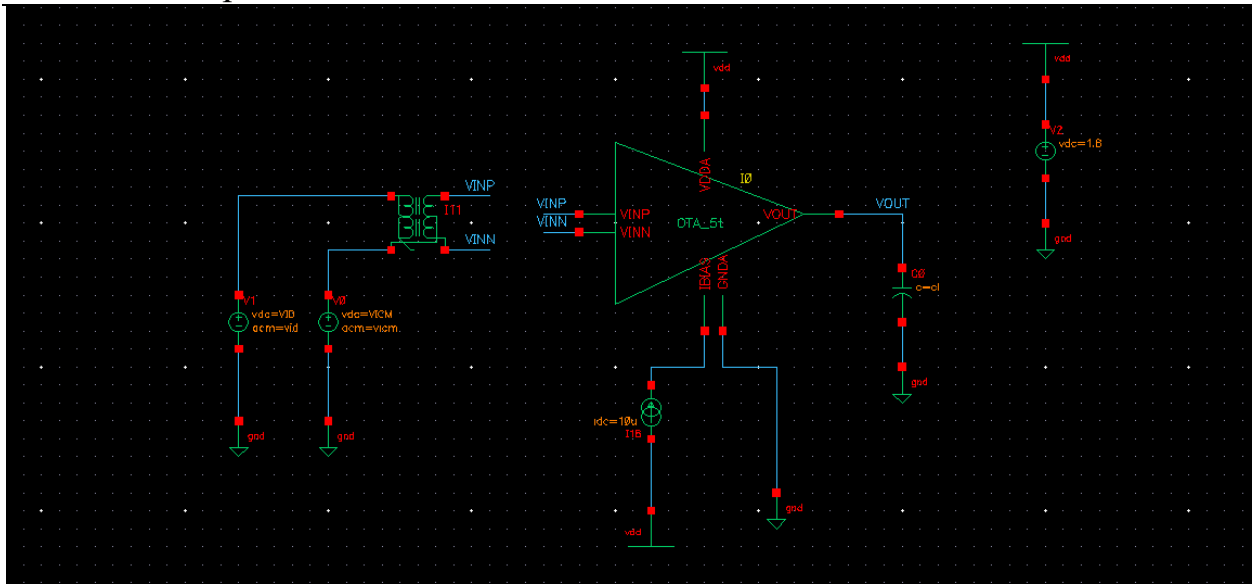


Figure 8:tb_for open loop gain

1)DC Operating Point and Bias Verification

After applying the calculated dimensions to the 5T OTA schematic, a DC analysis was performed to verify the biasing conditions and ensure all transistors operate in the proper region.

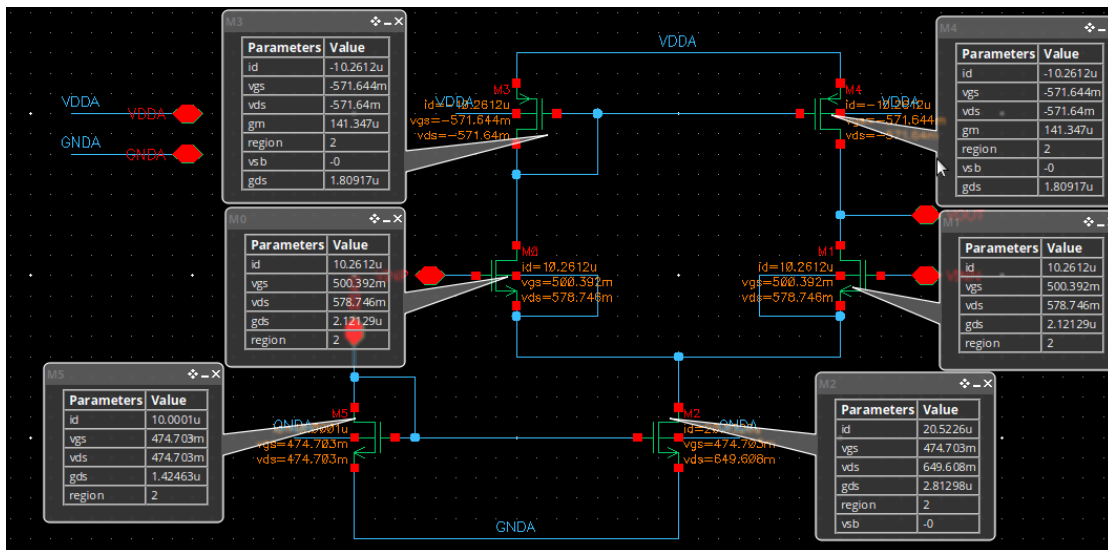


Figure 9:DC_OP



Commentary on DC Results:

- **Is the current (and g_m) in the input pair exactly equal?**

Yes, they are exactly equal.

Justification: As shown in the annotated schematic, both M0 and M1 carry an identical drain current of $I_D = 10.2612\mu A$. Since they have the exact same dimensions, bias conditions, and the differential input voltage is zero ($V_{id} = 0V$), their V_{GS} and V_{DS} are perfectly matched. Consequently, their transconductance (g_m) is also exactly equal. This happens because the simulation uses perfectly matched device models, resulting in zero systematic offset.

- **What is DC voltage at VOUT? Why?**

Value: The DC voltage at VOUT perfectly tracks and matches the DC voltage at the diode-connected node (the drain/gate of M3). Based on the properties of M4, $V_{DS4} = -571.644\text{ mV}$. Assuming $V_{DD} = 1.8V$, the DC voltage at VOUT is approximately 1.228 V.

Why: In an open-loop configuration with a perfectly matched circuit and a zero differential input, the circuit is completely symmetrical. To maintain this balance and force the exact same current ($10.2612\mu A$) through the right branch, the output node (drain of M4) must settle at the exact same DC potential as the corresponding node in the left branch (the diode-connected drain of M3)

2) Differential Small Signal Characteristics

To extract the differential open-loop parameters (DC Gain, Bandwidth, and GBW), an AC analysis was performed from 1Hz to 10GHz with $V_{id(AC)} = 1V$ and $V_{icm(AC)} = 0V$. The common-mode voltage (V_{ICM}) was set to the middle of the extracted CMIR.

A. Differential Gain vs. Frequency Plot:

The following plot shows the frequency response of the differential open-loop gain (in dB).

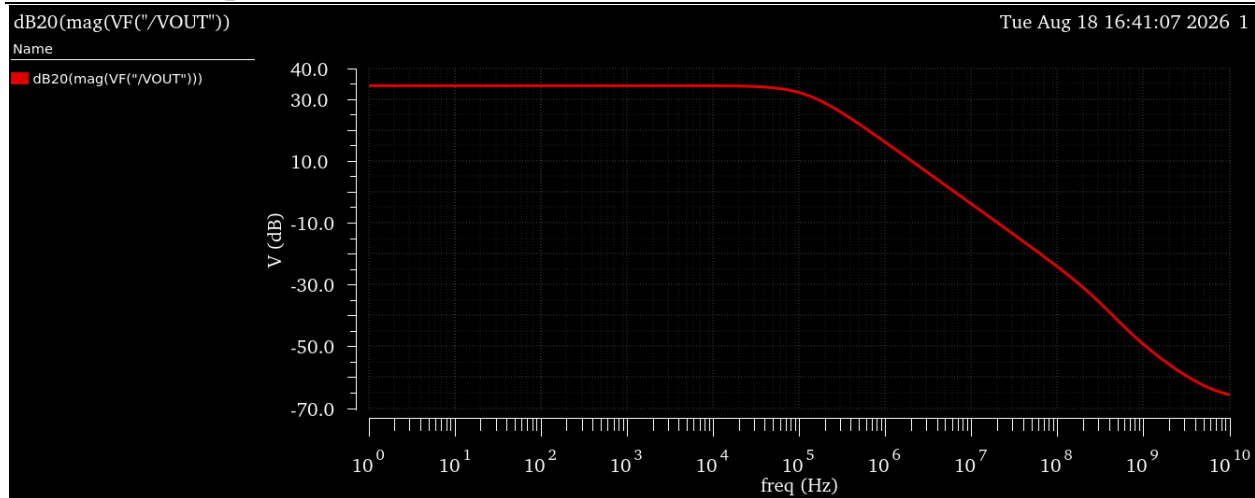


Figure 10: gain vs. frequency for diff small signal

B. Parameter Extraction and Comparison:

Using Cadence calculator expressions in the ADE XL environment, the key small-signal parameters were automatically extracted. The results are shown in the below:

Test	Output	Nominal	Spec	Weight	Pass/Fail
part2_mosfet:la_b_07_ttx1	dB20(mag(VF(*/VOUT*)))				
part2_mosfet:la_b_07_ttx1	Ao	52.71			
part2_mosfet:la_b_07_ttx1	A0_dB	34.44			
part2_mosfet:la_b_07_ttx1	BW	123.3k			
part2_mosfet:la_b_07_ttx1	fu	6.553M			
part2_mosfet:la_b_07_ttx1	GBW	6.501M			

Figure 11: Ao, BW, fu and GBW



The following table compares the target/hand-calculated values (from the initial design phase) with the simulated results:

Parameter	Hand Calculation / Target	Simulated Result (ADE XL)
DC Gain ($A_{0,dB}$)	≥ 34 dB (Target)	34.44 dB
Linear DC Gain (A_0)	~ 50 V/V	52.71 V/V
Bandwidth (BW)	124.77KHz	123.3 kHz
Gain-Bandwidth (GBW)	6.2385MHz	6.501 MHz

Commentary: The simulated results closely match the design targets. The DC gain of 34.44 dB slightly exceeds the minimum requirement of 34 dB. Furthermore, the simulated GBW of 6.501 MHz comfortably satisfies the target specification of ≥ 5 MHz, validating the accuracy of the g_m/I_D design methodology employed in Part 1.

3)Common-Mode Small Signal Characteristics

To evaluate the common-mode gain (A_{vcm}), an AC analysis was performed with $V_{icm(AC)} = 1V$ and $V_{id(AC)} = 0V$. The common-mode DC voltage (V_{ICM}) was set to the middle of the extracted CMIR.

A. CM Gain vs Frequency:

The frequency response of the common-mode gain was plotted using the AC analysis.

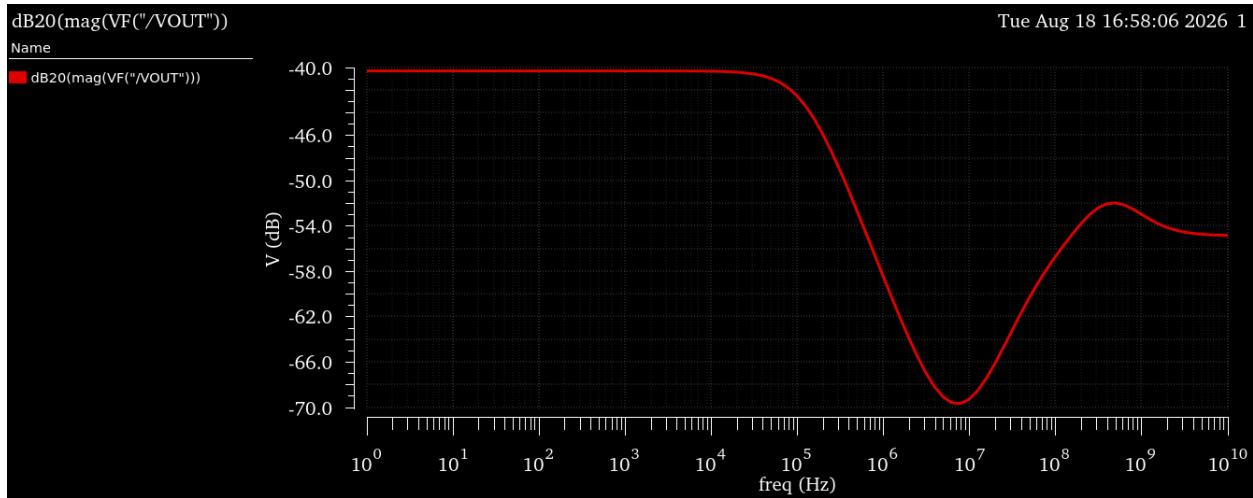


Figure 12: A_{vcm} vs. freq.

As shown in the plot, the low-frequency common-mode gain is approximately -40.2 dB.

B. Parameter Comparison: The following table compares the simulated common-mode gain with the hand calculations derived during the initial design phase.

C. A_{vcm} vs V_{ICM} : Following the Cadence Hint, an AC sweep of the design variable (V_{ICM}) from 0.8V to 1.5V at a low frequency (1 Hz) was performed to observe the variation of the DC common-mode gain across the input range.

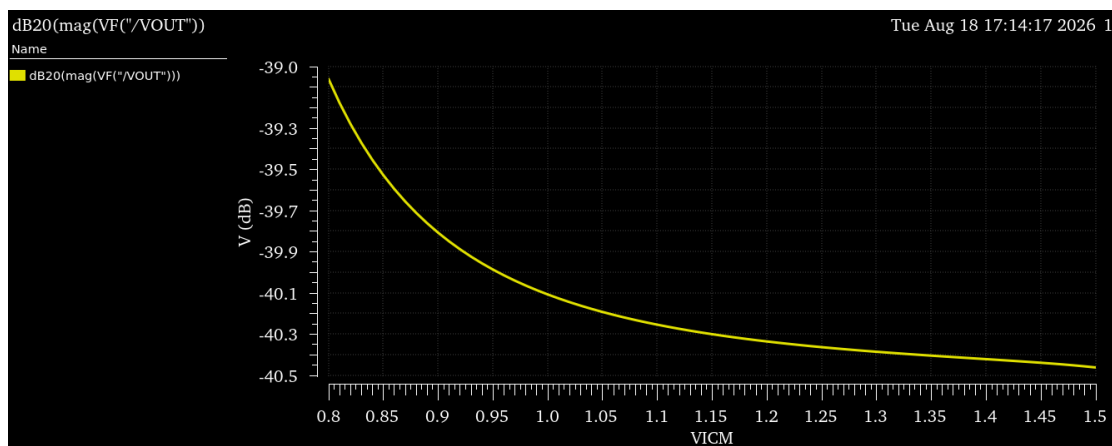


Figure 13: A_{vcm} vs. V_{icm}



The simulated plot shows the CM gain varying from approximately **-39 dB** at $V_{ICM} = 0.8V$ to **-40.5 dB** at $V_{ICM} = 1.5V$.

Justification for A_{vcm} Variation:

The common-mode gain magnitude is inversely proportional to the tail current source's output resistance (R_{SS}), approximated by:

$$|A_{vcm}| \approx \frac{1}{2g_{m3}R_{SS}}$$

As V_{ICM} increases towards **1.5V**, the drain-to-source voltage (V_{DS}) of the tail transistor (M5) increases, pushing it deeper into the saturation region. This increases its output resistance (R_{SS}), which consequently decreases the common-mode gain, improving the rejection. Conversely, as V_{ICM} decreases towards the lower limit (**0.8V**), M5 approaches the triode region. In this region, its R_{SS} significantly drops, causing the common-mode gain magnitude to increase (making it less negative in dB, approaching **-39 dB**).

4) Common-Mode Rejection Ratio (CMRR)

To accurately evaluate the CMRR without running two separate simulations, an XF (Transfer Function) analysis was utilized. The CMRR was extracted and plotted using the Cadence calculator expression provided in the lab instructions.

A. CMRR vs. Frequency: The following plot demonstrates the CMRR frequency response with the common-mode voltage set at the middle of the CMIR.

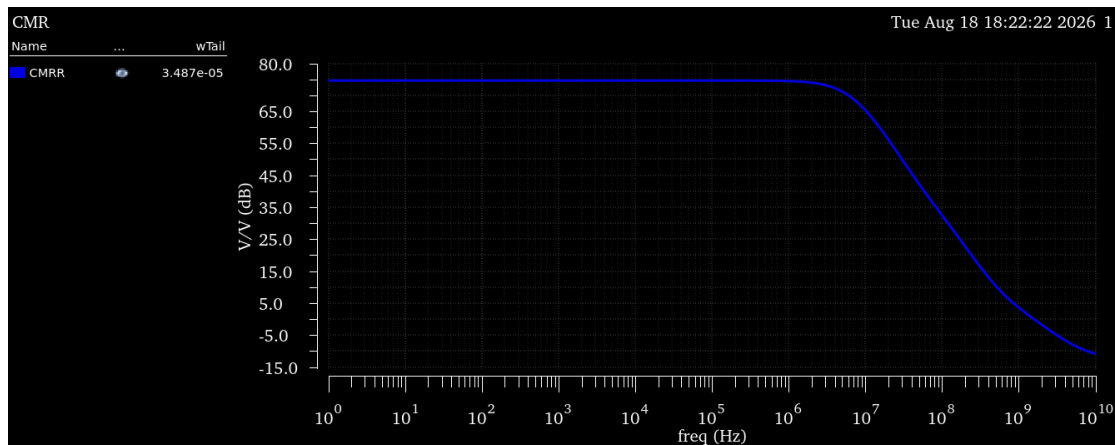


Figure 14: CMRR vs. freq



As observed from the simulation, the low-frequency CMRR is approximately **74.6 dB**.

B. CMRR vs. VICM (Optional Requirement): Following the Cadence hint, an AC sweep of the design variable (VICM) was performed at a low frequency (1 Hz) to observe the variation of the CMRR across the common-mode input range.

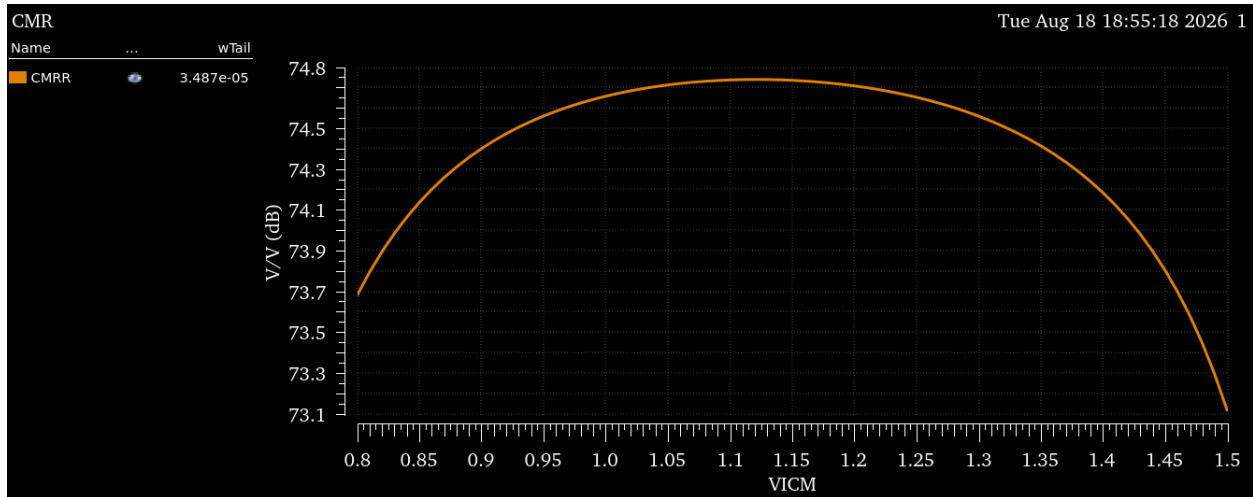


Figure 15: CMRR vs. VICM

The plot shows the CMRR peaking near the center of the CMIR and slightly degrading towards the edges, dropping to approximately 73.7 dB at 0.8V and 73.1 dB at 1.5V.

Justification for CMRR Variation:

The CMRR is fundamentally defined as the ratio of the differential gain to the common-mode gain:

$$CMRR = 20 \log \left| \frac{A_{vd}}{A_{vcm}} \right|$$

As justified in the previous section, when the input common-mode voltage ($VICM$) approaches the lower limit (0.8V), the tail transistor approaches the triode region. This decreases its output resistance (R_{SS}), which in turn causes the common-mode gain ($|A_{vcm}|$) to increase. Because the common-mode gain is in the denominator of



the CMRR equation, any increase in $|A_{vcm}|$ inevitably leads to a degradation (reduction) in the overall CMRR.

5) Differential Large Signal Characteristics

To analyze the large-signal behavior and the linear range of the OTA, a DC sweep of the differential input voltage (V_{ID}) was performed from -1.8V to 1.8V with a very fine step of 1mV . The common-mode voltage (V_{ICM}) was kept constant at the middle of the CMIR.

A. V_{OUT} vs. V_{ID} DC Transfer Characteristic:

The following plot illustrates the output voltage response to the differential input sweep.

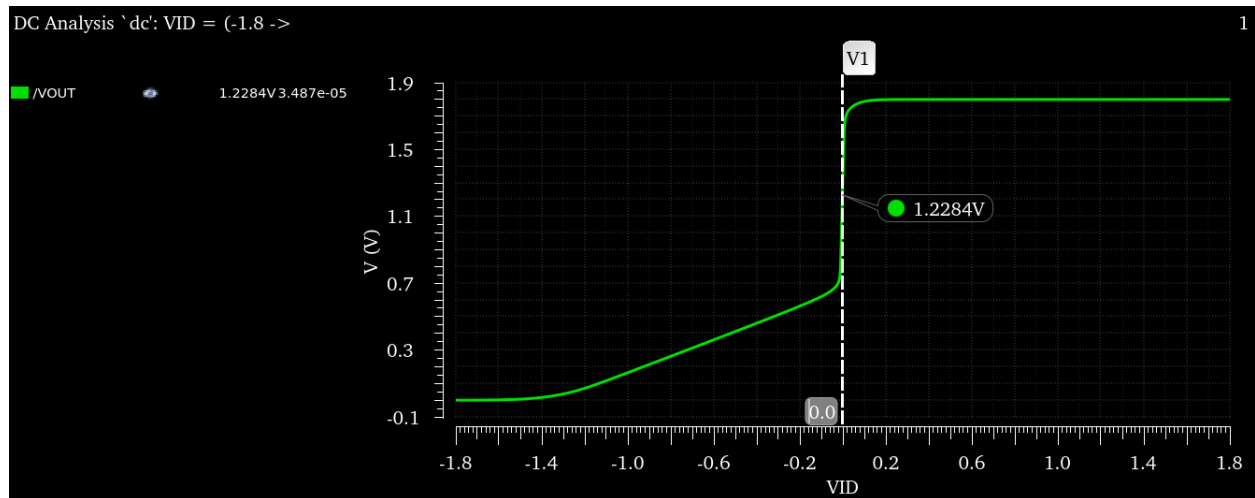


Figure 16: V_{out} vs. V_{id}

- **Value of V_{OUT} at $V_{ID} = 0\text{V}$:** As annotated on the plot, the output voltage is 1.2284 V .
- **Justification (Why?):** At exactly $V_{ID} = 0\text{V}$, the circuit is perfectly balanced and symmetrical. To maintain identical currents in both the left and right branches of the input pair, the output node (drain of M_4) must settle at the exact same DC potential as the corresponding node in the left branch (the diode-connected drain of M_3). This matches the DC operating point previously calculated.



B. Derivative of VOUT vs. VID:

To find the maximum gain region, the derivative of the VOUT vs. VID curve was plotted in dB.

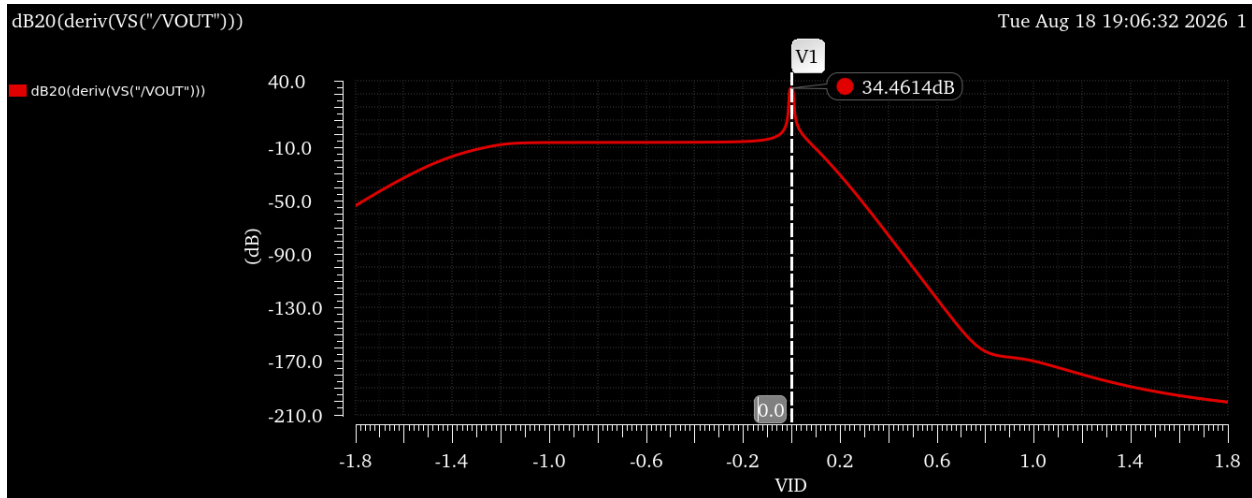
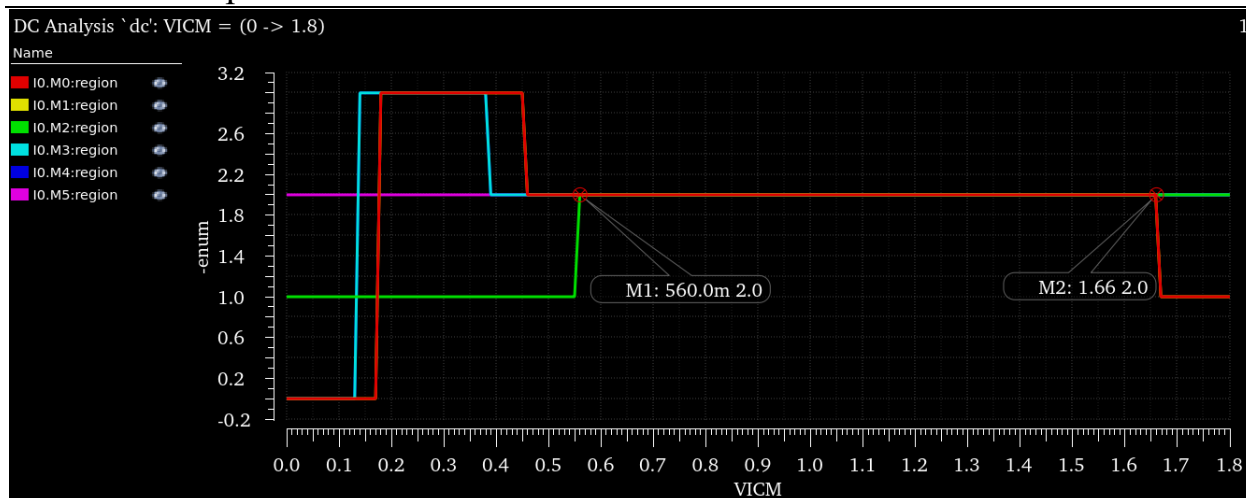


Figure 17: dV_{out}/dV_{id}

- **Peak Value and Comparison:** The peak of the derivative curve occurs exactly at $VID = 0V$ with a value of 34.46 dB. The derivative of the DC transfer characteristic at the quiescent point represents the small-signal DC gain. This peak value perfectly matches the open-loop differential gain ($A_{vd} = 34.44$ dB) obtained previously from the AC small-signal analysis, validating both simulation setups.

6) Common-Mode Large Signal Characteristics (Region vs. VICM)

To determine the Common-Mode Input Range (CMIR), a DC sweep of the common-mode voltage (VICM) was performed from 0V to VDD (1.8V). The operating "region" parameter for the input pair and the tail current source was plotted against VICM. In Cadence, Region 2 corresponds to the saturation region, which is required for proper OTA operation.



A. CMIR Extraction: The CMIR is strictly defined as the voltage range where both the input pair and the tail current source operate in saturation (Region = 2).

- **CMIR-low:** The lower limit is determined by the tail current source entering saturation. From the plotted markers, this transition occurs at **0.56V**.
- **CMIR-high:** The upper limit is determined by the input pair leaving the saturation region and entering the triode region. From the plotted markers, this transition occurs at **1.66V**.

B. Parameter Comparison: The following table compares the target design specifications with the simulated CMIR limits extracted from the region plot:

Parameter	Target Specification	Simulated Result
CMIR-low	$\leq 0.8V$	0.56V
CMIR-high	$\geq 1.5V$	1.66V

Commentary & Method Drawback: The simulated CMIR (0.56V to 1.66V) successfully covers and exceeds the required design specifications (0.8V to 1.5V). However, it is important to note a fundamental drawback of this specific extraction method: the "region" parameter is a simulator-internal variable. Therefore, this parameter cannot be experimentally measured in a physical lab environment.



PART 3: Closed-Loop OTA Simulation

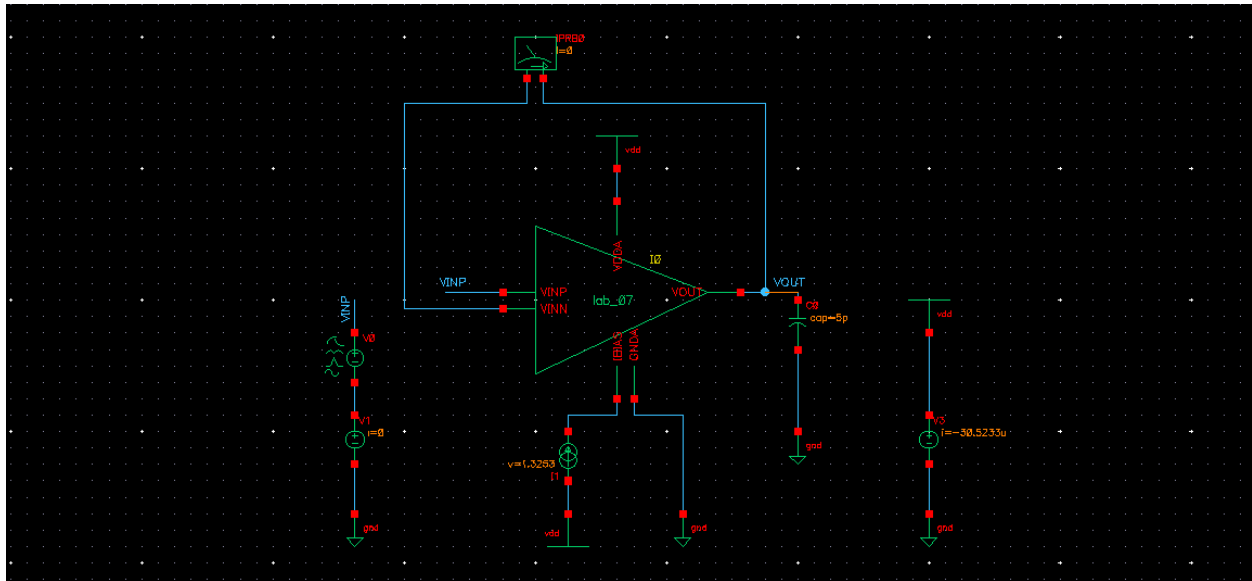


Figure 19: Closed-Loop OTA_TB

1) Closed-Loop Stability Analysis (STB)

A. DC Operating Point in Unity Gain Buffer Configuration

The OTA was configured as a unity-gain buffer using negative feedback. The input voltage was set near the lower edge of the common-mode range ($V_{IN} = \text{CMIR-low} + 50\text{mV}$).

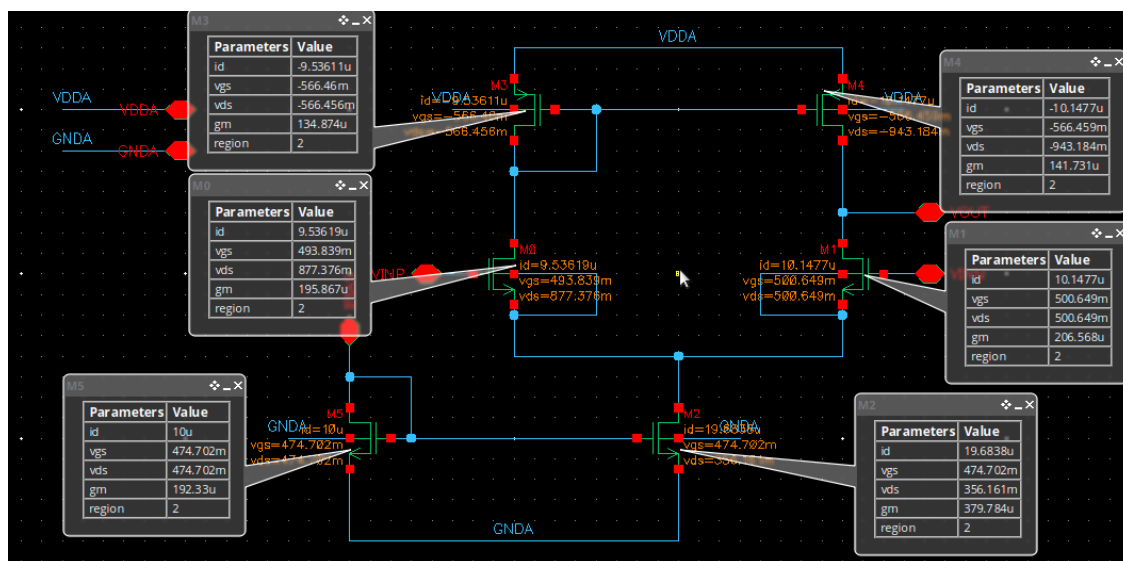


Figure 20: DC_OP_closed loop



- **Is the current (and gm) in the input pair exactly equal? Why?**

No, they are not exactly equal.

Justification: Because the OTA has a finite open-loop DC gain, the output voltage cannot perfectly track the input voltage. A small, non-zero differential input voltage (V_{id}) is required to drive the output node. This small differential voltage creates a slight imbalance in the gate-to-source voltages (V_{GS}) of the input pair transistors (M0 and M1). Consequently, this induces a systematic mismatch in their drain currents and transconductances.

- **Calculate the mismatch in ID and gm:**

Based on the annotated schematic parameters:

1. Drain Current Mismatch (ΔI_D):

- I_{D0} (Left branch) = 9.536 μA
- I_{D1} (Right branch) = 10.147 μA
- Absolute Mismatch (ΔI_D) = 10.147 - 9.536 = **0.611 μA**

2. Transconductance Mismatch (Δg_m):

- g_{m0} (Left branch) = 195.867 μS
- g_{m1} (Right branch) = 206.568 μS
- Absolute Mismatch (Δg_m) = 206.568 - 195.867 = **10.701 μS**

2) Loop Gain and Phase Margin Analysis (STB)

To evaluate the stability of the OTA in a closed-loop system, an STB analysis was performed from 1Hz to 10GHz with the OTA configured as a unity-gain buffer. The common-mode voltage was kept at the middle of the CMIR.

A. Loop Gain and Phase Plots:

The loop gain magnitude (in dB) and phase frequency responses were plotted using the direct plot form.

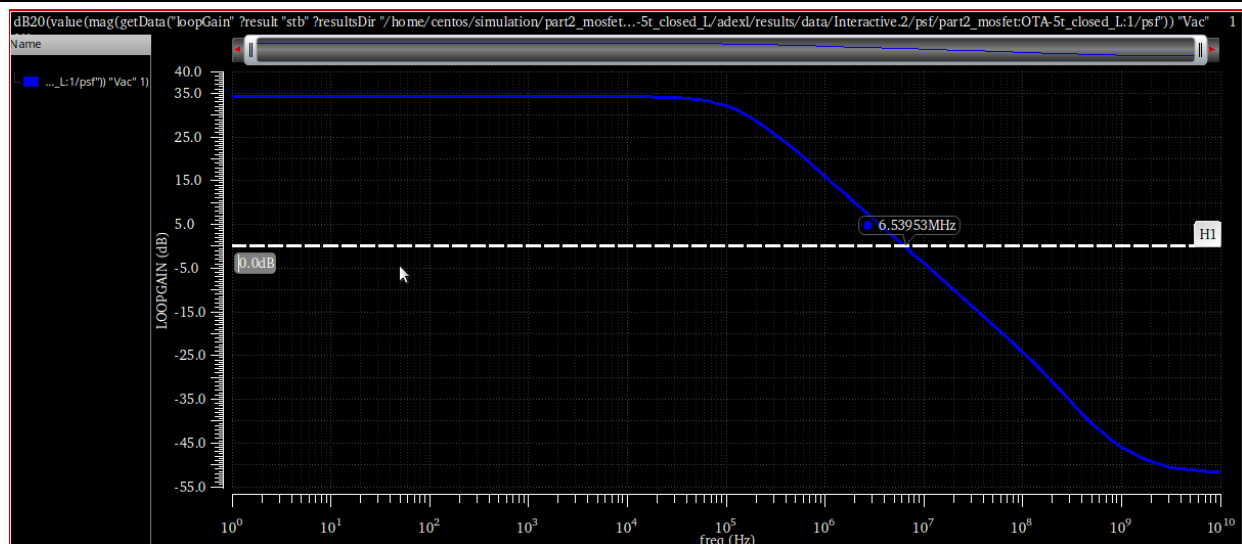


Figure 21: closed loop gain vs. freq.

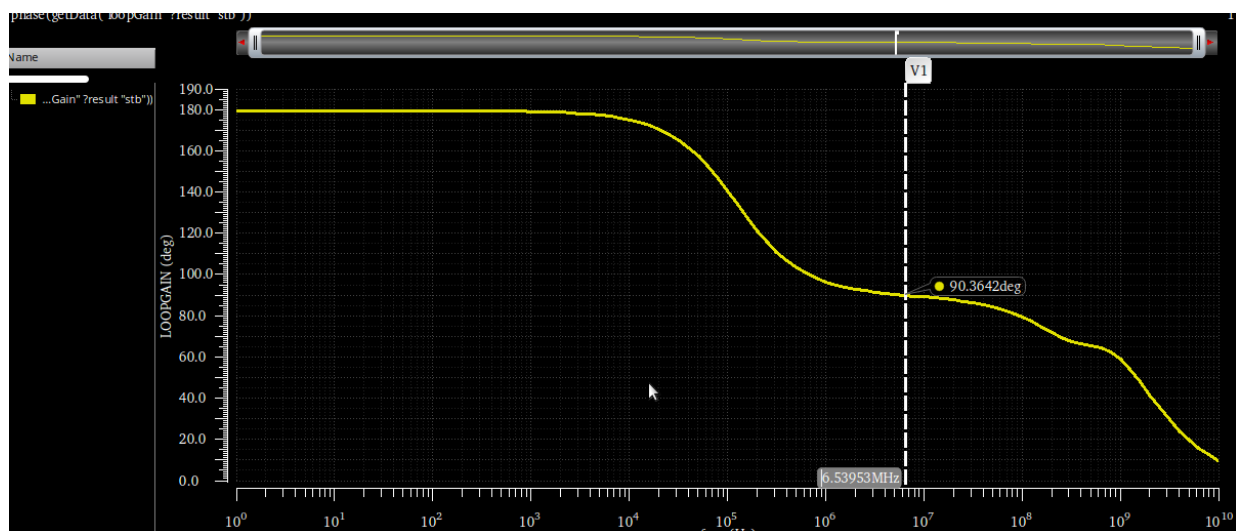


Figure 22: closed loop phase

As annotated on the plots, the Unity Loop Gain Frequency (0 dB crossing) occurs at **6.539 MHz**, and the Phase Margin at this frequency is **90.36°**



B. Open-Loop vs. Closed-Loop Comparison:

The following table compares the DC gain and GBW obtained from the previous open-loop AC simulation with the loop gain parameters extracted from the STB closed-loop simulation.

Parameter	Open-Loop (AC Analysis)	Closed-Loop (STB Analysis)
DC Gain	34.44 dB	≈ 34.44 dB
GBW / Unity Gain Freq	6.501 MHz	6.539 MHz

Commentary: The DC gain and GBW obtained from the STB analysis are virtually identical to those obtained from the open-loop AC analysis. The STB analysis calculates the Loop Gain (T). Since the OTA is connected in a unity-gain buffer configuration, the feedback factor is unity ($\beta = 1$). Therefore, the measured Loop Gain is exactly equal to the Open-Loop Gain ($T = A_{OL} \times \beta = A_{OL}$). The slight deviation in GBW (6.501 MHz vs 6.539 MHz) is negligible and is attributed to the slight DC operating point shift (mismatch) induced by closing the loop, as proven in the previous section.



C. Simulation Results vs. Design Targets:

Finally, the closed-loop AC characteristics are compared with the initial design specifications.

Parameter	Target Specification	Simulated Result (STB)
Loop DC Gain	≥ 34 dB	≈ 34.44 dB
GBW	≥ 5 MHz	6.539 MHz
Phase Margin	$\geq 70^\circ$	90.36°

Conclusion: All simulated parameters successfully meet or exceed the target specifications, confirming the validity of the g_m/I_D design methodology. The high phase margin (90.36°) indicates that the closed-loop system is highly stable with an excellent transient step response devoid of excessive ringing.